Robust Image Classification using Adversarial Defense

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This is to certify that the work present in this Project entitled **"Robust Image Classification using Adversarial Defense"** has been carried out by **[K Kavya Rishita , N Hima Manasa Lakshmi]** under my supervision. The work is genuine, original, and suitable for submission to the SRM University – AP for the award of Grades in Deep Learning :Methodologies and Techniques Course (DSC 507), Master of Technology of Technology in **School of Engineering and Sciences**.

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Abstract

In recent years, deep learning models have shown excellent performance in image classification tasks. However, one major issue is that these models can be easily fooled by very small changes in the input images, called adversarial attacks. These changes are so small that humans cannot even notice them, but they can cause the model to make completely wrong predictions.

In this project, we studied how these attacks affect model performance using datasets like MNIST, Fashion-MNIST, and SVHN. We applied two common attack techniques, FGSM and PGD, and observed that the model's accuracy drops significantly when these attacks are introduced.

To solve this problem, we used an autoencoder as a defense mechanism. The autoencoder helps in removing the noise added by the attacks and reconstructs a cleaner version of the image. When these cleaned images are given to the model, the accuracy improves compared to the attacked images.

Overall, this project shows both the weakness of deep learning models against adversarial attacks and how we can improve their robustness using simple defense techniques like autoencoders.

KEYWORDS: Deep Learning, Image Classification, Adversarial Attacks, FGSM, PGD, Autoencoder, CNN, ResNet, MNIST, Fashion-MNIST, SVHN, Model Robustness, Defense Mechanism

Statement of Contributions

1. Rishita (AP251220650010):

   She contributed by thoroughly studying and understanding the base research paper and its methodology. She contributed by working on the SVHN dataset, including its loading, preprocessing, and handling of real-world image complexities. She developed and implemented a ResNet model to improve performance on this complex dataset. Additionally, she implemented adversarial attack techniques such as FGSM and PGD to evaluate the robustness of the models. She also designed and applied an autoencoder-based defense mechanism to reduce the impact of these attacks and improve model performance.

2. Hima Manasa (AP25122060007):

   She was responsible for implementing the Convolutional Neural Network (CNN) model on the MNIST and Fashion-MNIST datasets. In addition, she handled data preprocessing tasks such as normalization, reshaping, and preparing the datasets for training. She also worked on visualizing the results by generating graphs including accuracy and loss curves, which helped in analyzing model performance. Furthermore, she assisted in drafting important sections of the report, particularly the methodology and results. Moreover, she prepared the architecture diagrams, including tListhe attack-defense pipeline, and was responsible for organizing and formatting the final project report.

Abbreviations

Abbreviation	Full Form
DL	Deep Learning
CNN	Convolutional Neural Network
ResNet	Residual Network
MNIST	Modified National Institute of Standards and Technology
SVHN	Street View House Numbers
FGSM	Fast Gradient Sign Method
PGD	Projected Gradient Descent
AE	Autoencoder
ReLU	Rectified Linear Unit
RGB	Red Green Blue
SGD	Stochastic Gradient Descent
Adam	Adaptive Moment Estimation
CE Loss	Cross-Entropy Loss
MSE	Mean Squared Error
Adv	Adversarial
Grad	Gradient
Img	Image

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Introduction

Nowadays, deep learning models are widely used for image classification tasks, and they perform very well in recognizing patterns from images. Models like Convolutional Neural Networks (CNNs) can easily identify handwritten digits, clothing items, and even real-world objects with high accuracy.

But even though these models are powerful, they have an important weakness. They can be easily fooled by something called adversarial attacks. In these attacks, very small changes are added to an image. These changes are so tiny that humans cannot notice them, but they can confuse the model and make it give wrong predictions.

In this project, we focus on understanding how these attacks affect model performance. We use two popular attack methods called FGSM and PGD. FGSM adds noise in a single step, while PGD is a stronger attack that adds noise multiple times to make the image more misleading. After applying these attacks, we observe how much the model's accuracy drops.

To solve this problem, we use an autoencoder as a defense technique. The main idea is to clean the attacked images before giving them to the model. The autoencoder learns how to remove the noise and reconstruct a better version of the image. When these cleaned images are used for prediction, the model performs better compared to when it directly sees the attacked images.

We test our approach on different datasets like MNIST, Fashion-MNIST, and SVHN. Through this, we show both how vulnerable the model is to attacks and how a simple defense method can improve its performance.

Overall, this project helps us understand the weaknesses of deep learning models and shows how we can make them more reliable in real-world situations.

1.1 Problem Statement

Deep learning models perform well in image classification but are vulnerable to adversarial attacks.Small, invisible changes in input images can mislead the model into making wrong predictions.This reduces the reliability of the model in real-world applications.Attacks like **FGSM** and **PGD** significantly affect model performance.There is a need for an effective method to defend against such attacks. This project focuses on using an autoencoder to reduce attack effects and improve robustness.

1.1.1 Dataset

The same CNN architecture is applied separately on all three datasets (MNIST, Fashion-MNIST, and SVHN). Each dataset is preprocessed according to its format, and the model is trained and evaluated individually. The results are then compared based on accuracy and loss.

1. MNIST Dataset

- Contains handwritten digits (0–9)
- Grayscale images (28×28)
- Very simple dataset

The MNIST dataset is used as a baseline dataset to test the basic performance of the CNN model.

2. Fashion-MNIST Dataset

- Images of clothing items
- Same size (28×28)
- More complex than MNIST

Fashion-MNIST is used to evaluate the model on slightly more complex image patterns compared to handwritten digits.

3. SVHN Dataset

- Real-world images of house numbers
- Colored images (32×32)
- More challenging

SVHN dataset is used to test the model's performance on real-world image data. Due to computational limitations and training time constraints, a subset of the SVHN dataset was used instead of the full dataset.

Methodology

Overview

This project is based on the implementation of a deep learning approach inspired by an existing research work on image classification using Convolutional Neural Networks (CNN). The original approach focuses on applying CNN models to standard benchmark datasets for classification tasks.

To enhance the work and introduce novelty, this project extends the implementation by incorporating multiple datasets, namely MNIST, Fashion-MNIST, and SVHN. While MNIST and Fashion-MNIST are relatively simple grayscale datasets, SVHN is a more complex real-world dataset containing colored images.

In addition to the basic CNN model, a more advanced deep learning architecture, ResNet (Residual Network), is introduced for the SVHN dataset to improve performance on complex image data. This modification helps in handling deeper networks effectively and addresses challenges such as vanishing gradients.

The same base CNN model is applied to MNIST and Fashion-MNIST datasets, while the enhanced ResNet-based approach is used for SVHN. This allows a comparative analysis of model performance across datasets with varying complexity, as well as evaluation of the impact of using advanced architectures.

2.1 Data Loading

In this project, three datasets were used: MNIST, Fashion-MNIST, and SVHN. Each dataset was loaded separately using Python libraries such as TensorFlow and NumPy.

- MNIST and Fashion-MNIST datasets were directly loaded from built-in TensorFlow datasets.
- SVHN dataset was loaded from external files.

Each dataset consists of training and testing data used for model development and evaluation.
Due to computational constraints, only a subset of the SVHN dataset was used to reduce training time while maintaining sufficient data for analysis.

2. Data Preprocessing

Before training the model, preprocessing steps were applied to prepare the data.

The following steps were performed:

- **Normalization:** Pixel values were scaled between 0 and 1
- **Reshaping:** Images were reshaped to include channel dimensions
- **Label Formatting:** Labels were converted into suitable format
- **Handling Differences:**
    - MNIST and Fashion-MNIST → grayscale images
    - SVHN → RGB images (handled accordingly)

These steps ensure uniformity and improve model performance.

3.Model Architecture

Two types of models were used in this project:

1. **CNN Model (for MNIST & Fashion-MNIST)**

The CNN model consists of:

- Convolutional layers
- ReLU activation
- Max-pooling layers
- Fully connected (Dense) layers
- Softmax output layer

This model is suitable for simple image classification tasks.

2. **ResNet Model (for SVHN)**

For the SVHN dataset, a **ResNet** architecture was used.

ResNet introduces:

- **Residual (skip) connections**
- Helps in training deeper networks
- Solves vanishing gradient problem

This makes it more effective for complex, real-world image data like SVHN.

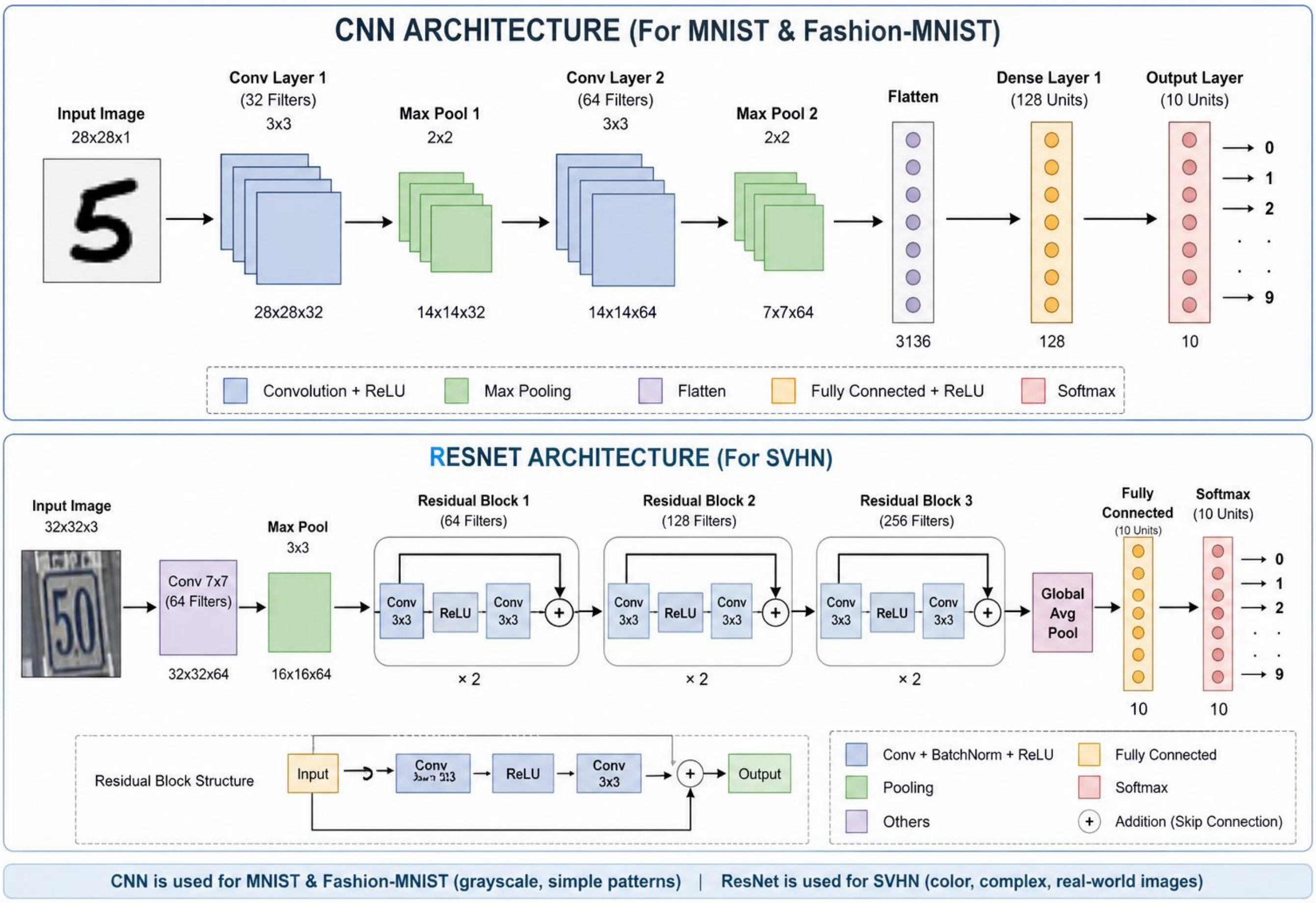


Fig 1 : Architecture of **CNN** for **MNIST & Fashion MNIST , RESNET** for **SVHN**

4. Model Training

The models were trained separately for each dataset using their respective training data.

Training configuration:

- **Loss Function:** Categorical Crossentropy
- **Optimizer:** Adam
- **Epochs**
- **Batch Size**

During training:

- The model learns features from images
- Loss decreases over time
- Accuracy improves gradually

CNN was trained on MNIST and Fashion-MNIST, while ResNet was trained on SVHN.

5. Adversarial Attack Generation

After training, the model is tested against adversarial attacks to check robustness.

FGSM Attack

- Adds small perturbations using gradient
- Fast but effective

PGD Attack

- Iterative version of FGSM
- Stronger and more harmful

These attacks slightly modify images in a way that:

- Humans cannot notice
- But model predictions become incorrect

6. Effect of Attacks

When adversarial images are given:

- Model accuracy drops significantly
- Shows vulnerability of deep learning models

This proves that models are not robust to small input changes.

7. Autoencoder-Based Defense

To solve this issue, an **Autoencoder (AE)** is used as a defense mechanism.

Working:

- Takes adversarial image as input
- Removes noise (perturbations)
- Reconstructs clean image

Structure:

- Encoder → compresses image

- Decoder → reconstructs image

The output image is closer to the original clean image.

8. Model Evaluation

After training, the models were evaluated using test datasets.

Evaluation metrics:

- **Accuracy**
- **Loss**

Results showed:

- MNIST → highest accuracy (simple dataset)
- Fashion-MNIST → moderate accuracy
- SVHN → lower accuracy due to complexity, but improved using ResNet

This comparison highlights the impact of dataset complexity and model selection on performance.

Dataset	Model	Clean	FGSM (0.05)	FGSM (0.1)	FGSM (0.15)	PGD (0.05)	PGD (0.1)	PGD (0.15)	Defense (0.05)	Defense (0.1)	Defense (0.15)
MNIST	VGG16	90.89	73.96	56.07	41.98	39.61	3.92	0.33	64.43	54.13	46.58
Fashion	VGG16	77.73	51.98	36.88	27.79	36.33	9.47	2.50	56.02	50.24	46.74
SVHN	ResNet18	83.75	10.80	1.55	0.55	3.50	0.10	0.00	22.80	7.00	2.25

Table 1 : Comparison of three datasets with different episilon values and their accuracies

8. Final Classification After Defense

The reconstructed (cleaned) images are again passed to the classifier (CNN/ResNet).

Results:

- Accuracy improves compared to attacked images
- Model becomes more robust

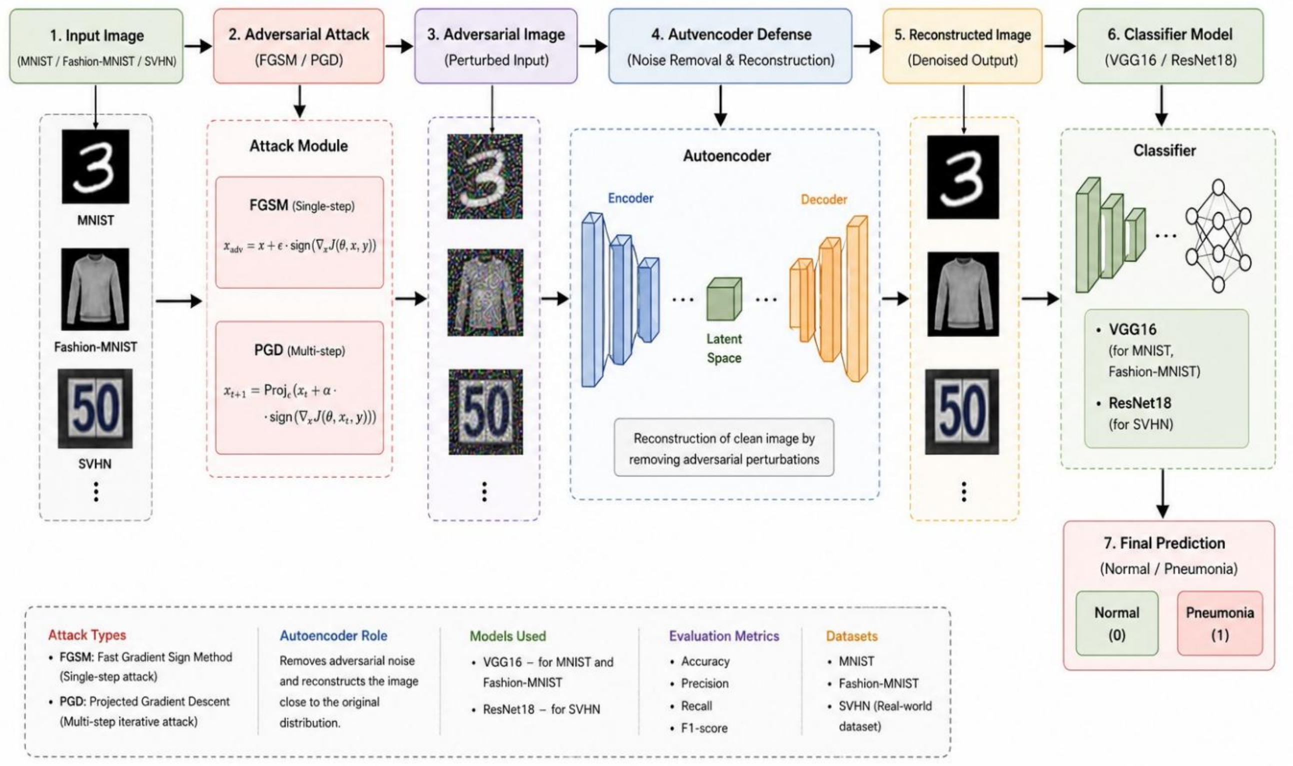
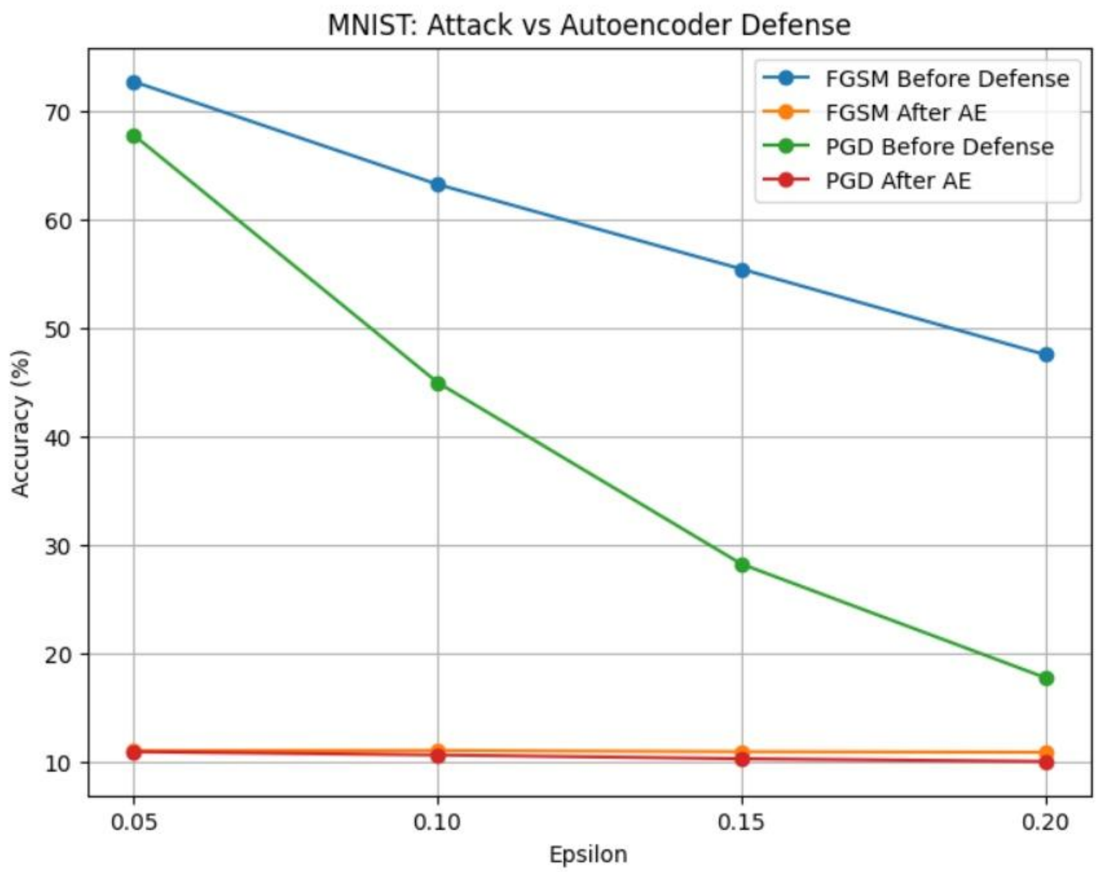
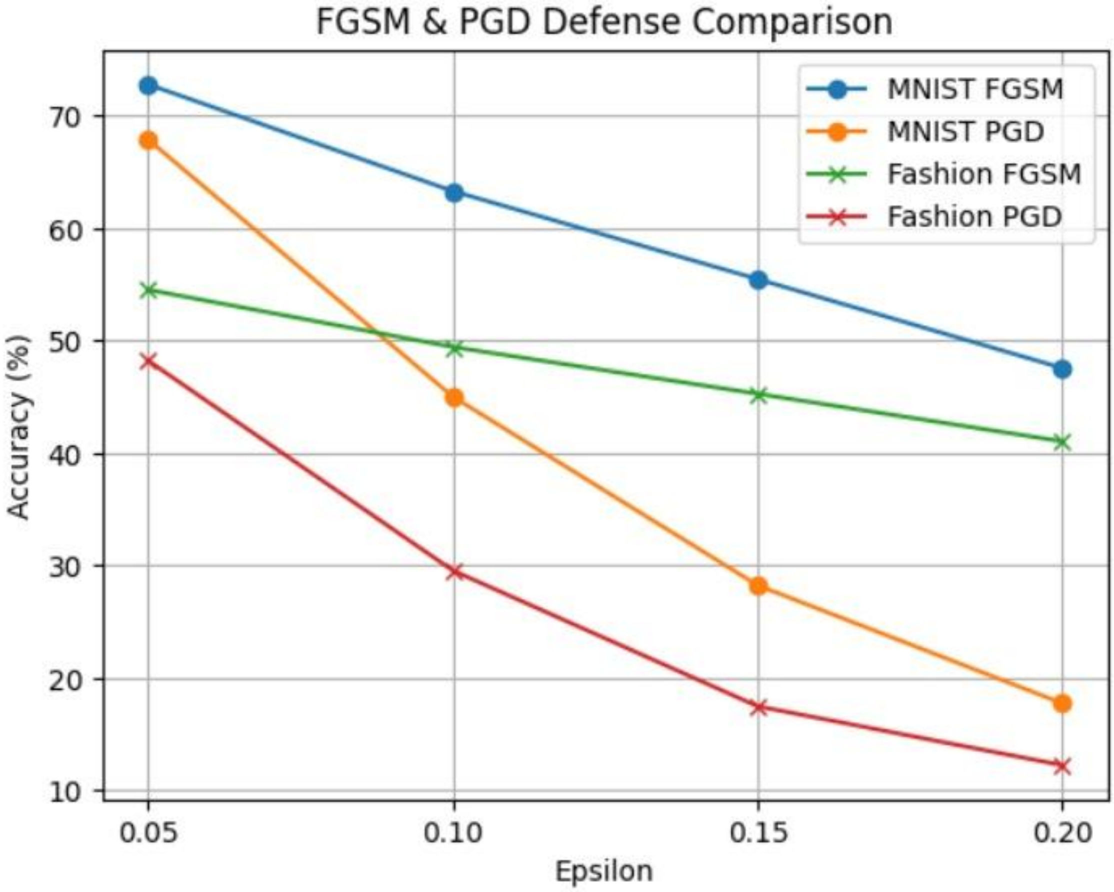


FIG 2 : Architecture of the Adverserial Defense Mechanism




FGSM & PGD Defense Comparison
MNIST FGSM
MNIST PGD
Fashion FGSM
Fashion PGD
Accuracy (%)
10
20
30
40
50
60
70
0.05
0.10
0.15
0.20
Epsilon

Concluding Remarks

This project successfully explored the performance and robustness of deep learning models in image classification tasks using MNIST, Fashion-MNIST, and SVHN datasets. The study demonstrated that while models such as CNN perform well on simpler datasets, their performance is significantly affected when exposed to adversarial attacks like FGSM and PGD.

The results clearly highlight the vulnerability of deep learning models to small, imperceptible perturbations in input data. However, the implementation of an autoencoder-based defense mechanism proved effective in reducing the impact of these attacks by reconstructing cleaner images and improving classification accuracy.

Additionally, the use of a more advanced architecture such as ResNet for the SVHN dataset helped in handling complex real-world data more efficiently. Overall, the project provides a comprehensive understanding of both the limitations and possible improvements in deep learning systems when dealing with adversarial scenarios.

This work emphasizes the importance of building robust and secure models, especially for real-world applications where reliability is critical.

Future Work

The current project can be extended in several directions to further improve model performance and robustness. One important improvement is to use the full SVHN dataset instead of a subset, which can provide better generalization and higher accuracy. Increasing the number of training epochs and performing hyperparameter tuning can also enhance the learning capability of the models.

More advanced deep learning architectures such as DenseNet, EfficientNet, or Vision Transformers can be explored to handle complex datasets more effectively. In addition, stronger adversarial attacks beyond FGSM and PGD can be implemented to evaluate the robustness of the models under more challenging conditions.

The defense mechanism can also be improved by using advanced techniques such as adversarial training, generative adversarial networks (GANs), or hybrid defense models instead of relying only on autoencoders. Data augmentation methods can be applied to increase dataset diversity and improve generalization.

Finally, this work can be extended to real-world applications such as medical image analysis, autonomous driving systems, or security-based systems, where robustness against adversarial attacks is highly critical.

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